

YASH CHAUHAN

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

EDUCATION

Guru Tegh Bahadur 4th Centenary Engineering College (GGSIPU) — New Delhi, India

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.5/10 | 2023 – 2027

CORE SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, C++

Frontend Development: React, Next.js, Redux, Tailwind CSS

Backend Development: Node.js, Express.js, REST APIs, WebSockets, Socket.io

Databases: PostgreSQL, MongoDB, Mongoose, Prisma ORM

Tools and Platforms: Git, Vercel, Render, Postman, Inngest, Resend API, Google Gemini AI

Core CS Concepts: Data Structures & Algorithms (DSA), Low-Level System Design, OOPS, Operating Systems, Computer Networks, DBMS

WORK EXPERIENCE

Software Engineering & Research Intern, DRDO – Scientific Analysis Group

June 2025 – August 2025

Scientific Analysis Group | New Delhi, India

- Engineered an automated end-to-end publishing platform using Python (Flask), Jinja2, and LaTeX, converting raw DOCX/PDF manuscripts into publication-ready, two-column IEEE conference and journal papers.
- Integrated local LLMs (Phi-3 Mini and Llama 3 via Ollama) to generate IEEE-standard titles, structured abstracts, and keywords, guaranteeing zero data leakage within an air-gapped defense environment.
- Architected a document parsing and image extraction pipeline using PyMuPDF and PDFMiner, extracting raw image byte streams via PDF xref tables and normalizing them into 300+ DPI RGB PNGs with dynamic column-aspect placement.
- Built an automated 3-pass pdflatex compilation engine via Python subprocesses, programmatically resolving .aux references, BibTeX citations, and floating double-column figure alignment with zero manual LaTeX intervention.
- Developed a split-screen reactive web editor with synchronized real-time PDF previews, hardening the platform with custom regex LaTeX-injection sanitization and bcrypt-secured authentication.

PROJECTS

EchoDesk — Enterprise B2B Support Platform and Embeddable Chatbot Engine

[Live](#) | [GitHub](#)

Next.js 15, Docker, MongoDB, Mongoose, Scalekit (SSO/Passkeys), Razorpay API, Google Gemini AI

- Built a multi-tenant B2B SaaS platform enabling custom chatbot deployment through a lightweight, zero-dependency client-side JavaScript widget.
- Developed a server-side knowledge-ingestion engine parsing uploaded PDFs and crawling external URLs into context-aware payloads for Gemini-powered chat responses.
- Integrated enterprise SSO and passwordless WebAuthn/Passkey authentication paired with Edge Middleware to enforce multi-tenant database isolation across organizations.
- Secured subscription billing using the Razorpay REST API and validated incoming event payloads with SHA-256 HMAC cryptographic signature verification.

FinSight — AI Financial Intelligence and Analytics Platform

[Live](#) | [GitHub](#)

Next.js 15, React 19, TypeScript, NextAuth.js, Prisma ORM v6, PostgreSQL (Neon), Inngest, Arcjet, Google Gemini AI

- Engineered a financial analytics application using Next.js 15 Server Actions, PostgreSQL on Neon, and Prisma ORM v6 with Zod schema validation on all data mutations.
- Implemented secure user authentication and session management with NextAuth, protecting all financial data routes behind authenticated server actions.
- Developed an automated Month-over-Month (MoM) anomaly detection engine flagging category-level spending spikes exceeding 30% relative to prior-month benchmarks.
- Implemented Inngest event-driven background queues to automate monthly budget burn-rate tracking and trigger email notifications through the Resend API.

ACHIEVEMENTS AND CERTIFICATIONS

- Solved 500+ problems on [Leetcode](#) with strong command of arrays, trees, graphs, dynamic programming, and sliding window techniques.
- Participated in 15+ hackathons with Top 5 finishes in 5+ events, including Top 3 in iQOO and Innov8 Hackathon, Top 8 at Build X: NSIT.
- Completed DSA Supreme 3.0 [certification](#), Code Help by Love Babbar, January 2026.